Unit: 6 Unit 8: Quadratics
Topic: Systems of Quadratic Equations

Name: ________________
Date: ___________

Homework (Jalapeno)

Q1. Solve the system: $x^2 + y^2 = 25$, $x^2 - y^2 = 9$
(A) $x = 4, y = 3$
(B) $x = 3, y = 4$
(C) $x = -4, y = -3$
(D) $x = -3, y = -4$
(E) $x = 2, y = 1$

Q2. Graph the system: $y = x^2 + 2$, $y = -x^2 + 4$
(A) Two intersecting parabolas
(B) One parabola and one line
(C) Two parallel lines
(D) One circle and one line
(E) Two intersecting lines

Q3. Solve the system: $x^2 + 2y^2 = 12$, $x^2 - y^2 = 3$
(A) $x = 2, y = 1$
(B) $x = -2, y = -1$
(C) $x = 3, y = 2$
(D) $x = -3, y = -2$
(E) $x = 1, y = 2$

Q4. In ancient Babylon, a farmer had two quadratic equations to model his crops: $y = x^2 + 3$, $y = -x^2 + 5$. What is the point of intersection?
(A) (1, 4)
(B) (-1, 4)
(C) (0, 3)
(D) (1, 5)
(E) (0, 0)

Q5. Solve the system: $x^2 - y^2 = 7$, $x^2 + y^2 = 11$
(A) $x = 2, y = 1$
(B) $x = -2, y = -1$
(C) $x = 3, y = 2$
(D) $x = -3, y = -2$
(E) $x = 1, y = 3$

Q6. The ancient Egyptians used quadratic equations to build the pyramids. If $y = x^2 + 2$ and $y = -x^2 + 6$, what are the solutions?
(A) $x = 1, y = 3$
(B) $x = -1, y = 3$
(C) $x = 2, y = 6$
(D) $x = -2, y = 6$
(E) $x = 0, y = 2$

Q7. Solve the system: $x^2 + y^2 = 20$, $x^2 - y^2 = 4$
(A) $x = 2, y = 2$
(B) $x = -2, y = -2$
(C) $x = 3, y = 1$
(D) $x = -3, y = -1$
(E) $x = 1, y = 4$

Q8. The ancient Greeks used quadratic equations to model the motion of objects. If $y = x^2 + 1$ and $y = -x^2 + 3$, what is the point of intersection?
(A) (0, 1)
(B) (0, 3)
(C) (1, 2)
(D) (-1, 2)
(E) (0, 2)

Open Ended Questions (FRQ)

Q9. Solve the system: $x^2 + 2y^2 = 15$, $x^2 - y^2 = 5$. Show your work and explain your reasoning.

Q10. The ancient Chinese used quadratic equations to model population growth. If $y = x^2 + 2x + 1$ and $y = -x^2 + 4x - 3$, what are the solutions? Use graphical and algebraic methods to solve the system.

Chef's Tips

- Tip 1: Check for extraneous solutions
- Tip 2: Use graphing to visualize the solutions
- Tip 3: Be careful with signs and coefficients